

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

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Investment Banks (Chapter 22)



Preview of talk

- ✓ Smooth functioning of securities markets requires Financial Institutions
- ✓ Many types of FI's (chapter 22)
 - **Investment Banks**
 - Brokers, Venture Capitalists, Private Equity
- ✓ We will only talk about IBs
 - IBs firms help buy/sell stocks, raise financing through equity and bonds issues etc...
 - other types not part of the exam
- ✓ On IB's
 - history
 - the underwriting process
 - M&A's
 - IPO's

Investment Banks

✓ **Investment banks (IBs)** are financial intermediaries that *mostly*:

1. Underwriting / placement of securities

- at IPOs
- they intermediate funds between the issuers of securities (debt, equity) and investors

2. Deal maker in M&A's and spin-offs

3. Private broker/bank to the very wealthy

✓ In doing 1-3, they “*research*” companies (provide analysts' forecasts)

Investment Banks

✓ Differences with CBs:

- IBs do **not** collect deposits and make loans (*asset transformation*) like commercial banks do ...
- ... but they
intermediate the flow of funds
in primary & secondary markets

✓ Their revenues? **FEES!**

- charged to clients for their services, usually not commissions on stock/bond trades

*For a long time, they were the elite of Wall Street.
Landscape has changed dramatically after 2008!*

Investment Banks Pre and Post Crisis

✓ IBs in 2006

1. **Lehman (L)**
2. **Bear Stearns (BS)**
3. **Morgan Stanley (MS)**
4. **Merrill Lynch (ML)**
5. **Goldman Sachs (GS)**

Investment Banks Pre and Post Crisis

- ✓ IBs in 2006
 1. Lehman (L)
 2. Bear Stearns (BS)
 3. Morgan Stanley (MS)
 4. Merrill Lynch (ML)
 5. Goldman Sachs (GS)

- ✓ IB in 2009 (but filed for *Commercial Bank charter*)
 1. **GS**
 2. **MS**
 - BS was acquired by JP Morgan
 - ML by Bank of America
 - L failed

- ✓ Technically no IB left ...
 - they are all Commercial Banks, although GS and MS retain a big 'investment banking' arm

Investment Banks History

- ✓ Before 1933, IB were part of “Universal” banks, but during Great Depression 10,000 banks failed (about 40%)
 - allegedly because many had given loans for the purchase of securities
 - sell loans to customers to buy overpriced shares... if price falls (as in a crisis) customer has loss on shares, bank too on loan!
- ✓ IBs were created by the *Glass-Steagall Act (1933)*
 - **idea:** **separate** ‘normal’ banking activity (deposits ⇒ loans) ...
 - ... from other risky activities (trading/underwriting of securities)
- ✓ *Glass-Steagall Act*
 - made it **illegal** for Commercial Banks (CB) to buy or sell securities on behalf of its customers
 - separation of commercial banks (CB) and IB avoids conflicts of interests that create potential risks of financial stability (sell loans to buy overpriced shares)

Investment Banks History

- ✓ In 1999 the *Glass-Steagall* was **repealed** by the *Gramm-Leach-Bliley Act* (under President Clinton)
 - some say this was one contributing factor to the crisis
 - since then, the line between CBs and IBs has become very **blurry**

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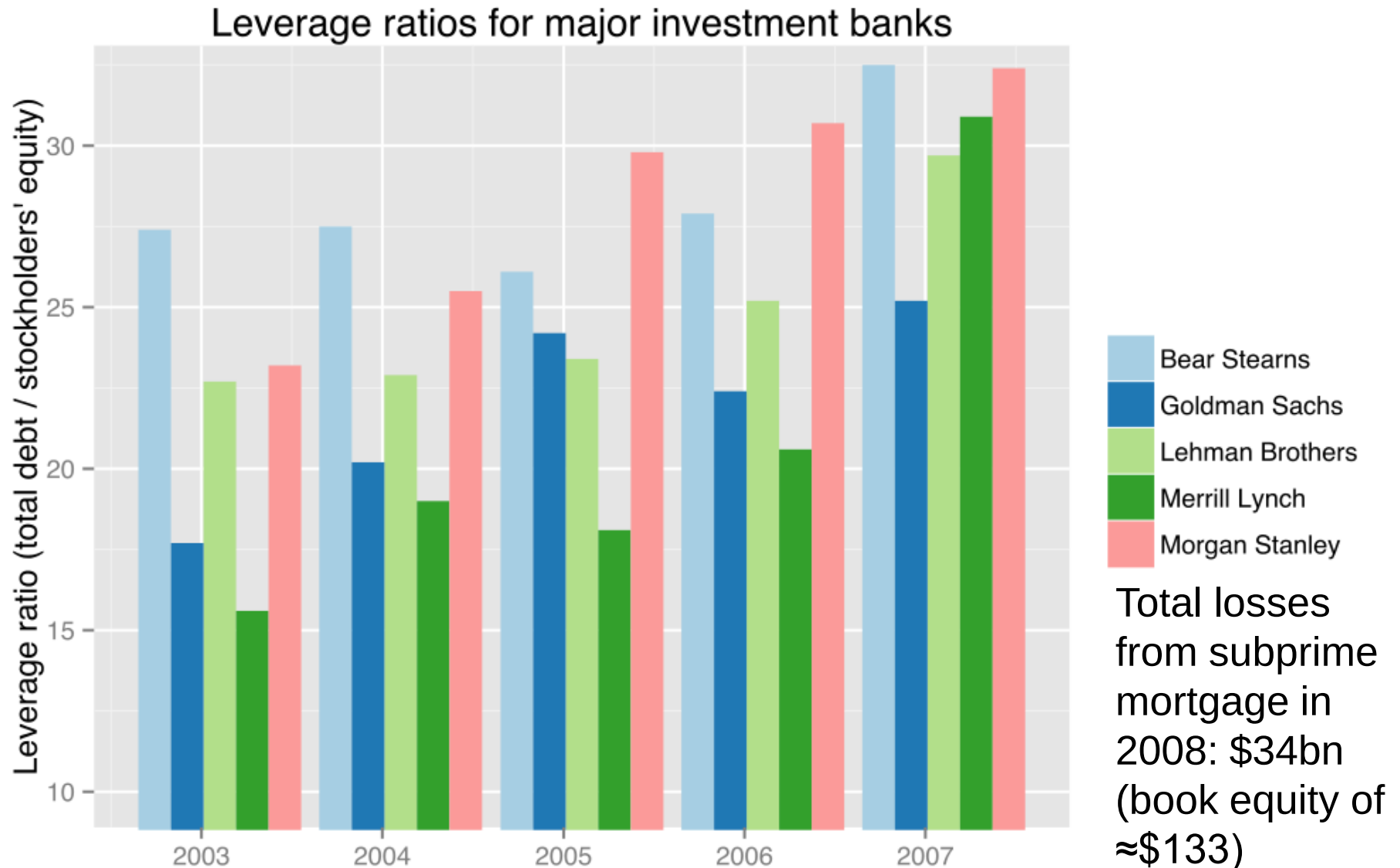
- ✓ In 2010 the *Dodd-Frank Act* (President Obama) partly **reintroduced** some separation between CB and IB
 - *Volcker Rule*: prohibits “proprietary trading” at CB
 - bank cannot trade for profit with its own money (only on behalf of clients)
 - again the logic is *separate* traditional banking from riskier activities
 - does this mean that GS (now a CB) cannot do trading anymore? Well, not really

- ✓ In 2018 President Trump **eased** some of the rules

In Europe

- ✓ No distinction between IB and CB: «**Universal banks**»
 - Commercial Banks could become IB by expanding into investment banking activities (underwriting, advising...)
- ✓ For example, Deutsche Bank did so in the 1990s
- ✓ Major universal banks (both IB and CB) in Europe:
 - DB, Commerzbank (DE)
 - UBS, CS (CH)
 - BNP Paribas, Credit Agricole, SocGen, Natixis (FR)
 - Santander, BBVA (ES)
 - UniCredit, Intesa San Paolo (IT)
 - RBS, Barclays, HSBC, Lloyds (UK)
 - ABN AMRO, ING, Rabobank (NL)

Investment Banks Riskiness



Underwriting Stocks and Bonds

- ✓ Main traditional business area of IB is to help firms
 - *get listed*
 - and
 - *issue bonds*
- ✓ Why would a firm need this service, and not issue securities on their own? Because
 1. firms don't do it often and there are **high transaction costs**
 2. IBs give **advice** on
 - *when* to issue
 - *what* to issue
 - *what price*
 - they also know *potential buyers* and *procedures*
 3. provide **implicit guarantee** about the issuance quality (*asymmetric info*)

Underwriting Stocks and Bonds:

1. Transaction Costs

High transaction costs

- ✓ investment banks have massive expertise in emitting new securities ...
- ✓ ... whereas many firms lack the necessary knowledge because they rarely access primary markets ⇒ lower transaction cost

- ✓ Indeed,
 - rules are complicated (see for example SEC registration statement for [Facebook, Inc.](#))
 - inaccurate registration statements can result in *lawsuits*
 - debt issues: require a credit ratings from a rating agency
 - equity issues; decide:
 - where (stock exchange/OTC) the stock will be eventually traded
 - when it will be issued, etc...

Underwriting Stocks and Bonds:

2. Giving advice

IB also can **advise** on type, price and date of issuance

✓ **What** type (stocks or bonds) to issue?

Q: if PE industry ratios are high and interest rates are high, should you issue stocks or bonds?

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A: stocks (high P_S) rather than bonds (low P_B , bcs i_B is high)

✓ **When** to issue?

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Explain current market conditions

At **what price**?

- For a *Seasoned Equity Offering (SEO)*

- For an *Initial Public Offering (IPO)*

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Explain current market conditions

At **what price?**

- For a *Seasoned Equity Offering (SEO)*
 - IB can look at current price in the secondary market
- For an *Initial Public Offering (IPO)*
 - not so easy

Underwriting Stocks and Bonds:

3. Asymmetric Information

The IB also help solve an **asymmetric information** problem when a firm issues stocks/bonds:

- ✓ investors don't know the true value of the firm ...
 - they could **research** the company, but it is costly
 - they do not trust the company itself to reveal honest and accurate information
- ✓ ... but they **trust** the IB (reputation)
- ✓ the firm contacts the IB about the *initial interest* in the issue of a particular security
- ✓ if the IB says yes, it is implicitly **guaranteeing** the quality of the issue to the public.
 - «implicitly» because of 2 methods of underwriting (**next**)

Underwriting Stocks and Bonds:

3. Asymmetric Information

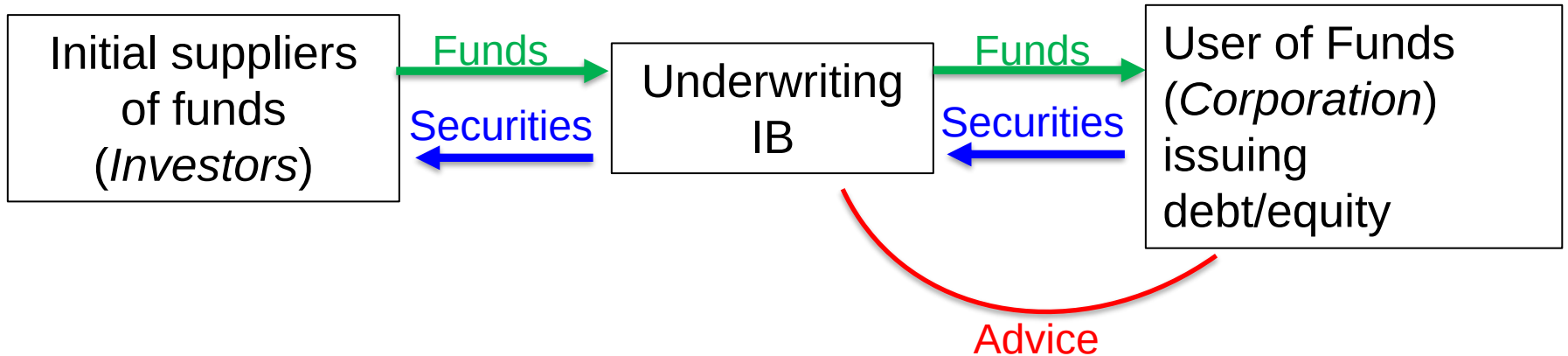
So IBs have **reputational capital** at stake

- ✓ It is a *repeat business*, so their income also depends on customer returning and on investors' trusting the IB judgement
- ✓ The IB is incentivized to do ***due diligence*** on the issues they distribute to the public
- ✓ If IB lose investors' confidence, their ability to make deals is *severely impaired*

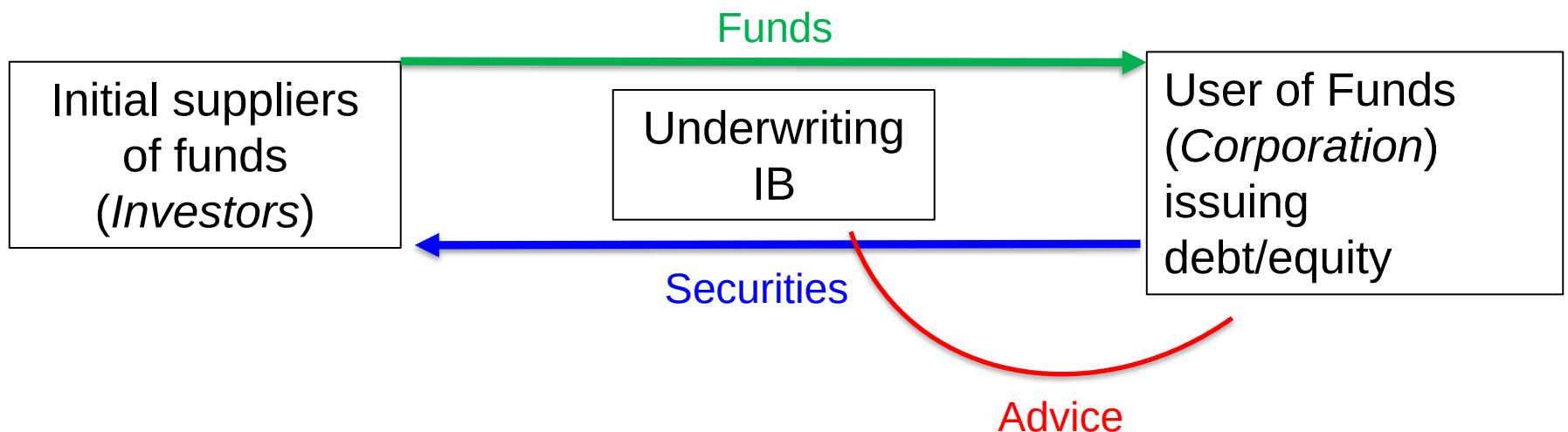
The Underwriting Process

Two main types of underwriting

Firm Commitment



Best effort



The Underwriting Process

Two main types of underwriting

1. Traditional (Firm Commitment) Underwriting

*IB buys **full** issue at a **pre-determined** price and then distributes the issue*

- the firm is *guaranteed* the price for its shares
- the IB profit is the difference between
 - the price paid to the issuer (**purchase price**)
 - and the price received in the secondary market (**offer price**)
- since this is quite risky for IB, usually IB ask other IBs to join and form a **syndicate**
 - EX. for Facebook IPO the syndicate was: MS, GS, BofA ML, JP Morgan, Allen & Company

The Underwriting Process

2. Best-effort underwriting

no price guarantee, **only distribution**

- the firm is **not** guaranteed a price because the IB does **not** purchase the entire issue (**nor** guaranteed that the issue can be sold entirely)
- the IB gets a commission for selling securities
 - lower than fees with firm commitment
- the IB has less incentives to do due diligence because it is not putting its reputation at risk
- in best-effort underwriting the IB is no longer certifying the issuance quality
 - not good for firms with substantial amount of asymmetric information (better choose traditional underwriting)

The Underwriting Process

Traditional (Firm Commitment)

Vs

Best-effort

Q1: best-effort underwriting is mostly used for IPO or SEOs?

Q2: which method do you think is better for small or less known firms?

Q3: which method creates more conflict of interest over price?

The Underwriting Process

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A1: **SEO**. There is a price already on the secondary market and Asy
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A2: **traditional** underwriting is better: with this method IB certifies the issuance quality, and thus reduces Asy Info for firms where these problems are more relevant

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Q3: which method creates more conflict of interest over price?

A3: with **traditional** method IB wants to buy shares underpriced (because the risk of not reselling at appropriate offer price is on IB) and re-sell them overpriced

The Underwriting Process

- ✓ The goal of underwriting is to **fully subscribe** (all of the shares have been spoken for)
- ✓ Both cases below are not good outcomes for the issuing company and for the IB
 - 1. Undersubscribed:** underwriting syndicate unable to generate interest in all of the available shares
 - IBs can lose huge amounts of money: need resell at a **lower price ...**
 - 2. Oversubscribed:** interest in more shares than are available
 - not good either: IB should have set a **higher price**
 - issuing firm may feel uneasy about having received less than its fair value
 - IB's reputation as incompetent underwriter

The Underwriting Process

Private Placements

- the entire issue is sold to a *small, select group of investors*
- ✓ Advantage is that there is no need for SEC registration
- ✓ Need buyers with *large pockets*:
 - insurance companies, commercial banks, mutual funds...
- ✓ IB facilitates transaction also by *identifying* potential purchasers
- ✓ Securities:
 - *mostly* private placement of *bonds*
 - *rarely* done with equity issues
- ✓ GS most active IB in private placements

Why getting listed?



The New York Times

Account ▾

Inside Airbnb, Employees Eager for Big Payouts Pushed It to Go Public

Tension has grown among a 6,000-person work force as it waits to sell company shares, people with knowledge of the situation said.

Underwriting at IB under threat?

Spotify's Splashy Debut Pressures Banks

Shares of the music-streaming service start with a flourish then steadily wear down, but they still finished above their trading price in the private markets



A specialist announces the start of trading in Spotify at the New York Stock Exchange on Tuesday. PHOTO: JUSTIN I ANE/FPA/SHUTTERSTOCK

*"[...] To go public, Spotify executed an unusual move called a **direct listing**, forgoing investment-banking underwriters and opting not to raise any money for itself. [...]*

*The company paid about **\$36 million in total fees***

*Snap was similar in size during its March 2017 offering, it paid nearly **\$100 million** to its large team of underwriters"*

Mergers and Acquisitions (M&A)

- ✓ Another traditional business area for IB: **M&A deal making**
 - first, IB determines the *value* of the company
 - then, IB makes *discreet inquiries* to find potential buyers/targets
- ✓ A **Merger** is the combination of two firms into one
 - management at both firms agree
 - shareholders simply *convert their stocks* into the new firm stocks
 - IB give their support to this process
- ✓ **Acquisition** one firm takes over another by buying its stocks
 - this can be a **friendly** takeover. Often time a company in financial distress will seek out a buyer
 - deal may be a **hostile** takeover, where the target resists (does not wish to be acquired)
 - acquirer tries to buy shares to have majority in board of directors
 - in *both* cases, IB helps both (acquirers and targets)

Conflicts of Interest

Conflict of interest is a type of *moral hazard* that arises when there are multiple interests, and serving one interest is detrimental to the other

- ✓ IB research (*i.e.* provide analysis on) companies they also help underwrite

Q1: where is the conflict of interest? What are the multiple interests the IB is serving?

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Conflict of interest is a type of *moral hazard* that arises when there are multiple interests, and serving one interest is detrimental to the other

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Q1: where is the conflict of interest? What are the multiple interests the IB is serving?

A: There are two party's interest: *investors* and the *company*

- the public (investors) want an unbiased review of the company: they benefit from an *realistic* review
- the company benefits from an *optimistic* review

Serving one (say, investors) damages the other (company)

Conflicts of Interest

- ✓ Traditionally IB would favor the issuing firm
 - MS internal memo, leaked to the *WSJ* on July 1992
 - “...adopt a policy, fully understood by the entire firm, including the Research Department, that **we do not make negative or controversial comments** about our clients as a matter of sound business practice”*

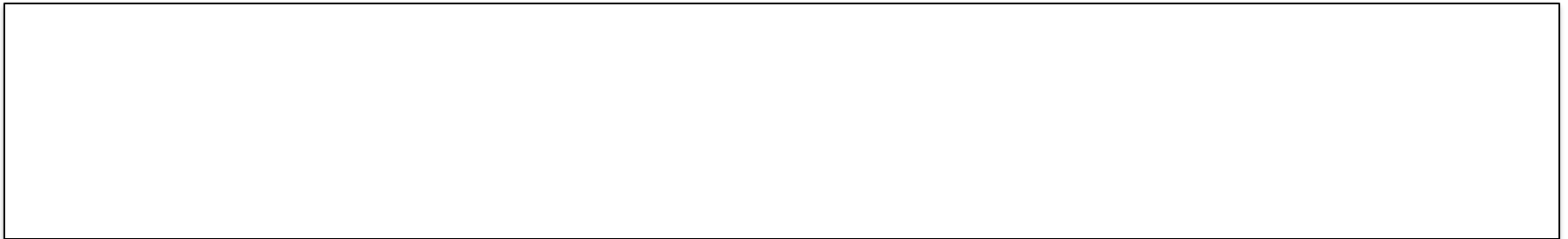
- ✓ Another common practice was that of **spinning**
 - first, IB places *underpriced* shares of an IPO to another company executives, in exchange for future business with the IB
 - then, executives of another company could profit by re-selling shares at a higher price on secondary market
 - IB loses «today», but gains business with the company's future SEOs
 - Nowadays spinning prohibited

Pricing the IPO

- ✓ Pricing the IPO is technically hard (no market price yet!) but also inherent tension (**conflict of interest**) between
 - *IB profits*
 - and
 - *the interest of the company*

Indeed:

1. On the one hand, the IB wants to buy the issue at the *lowest possible price* and then sell it at a *highest possible price* to the public



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Indeed:

1. On the one hand, the IB wants to buy the issue at the *lowest possible price* and then sell it at a *highest possible price* to the public
2. On the other hand, they need to ensure that the issuer is *satisfied* with the *price*
 - issuer cares also of having fully subscribed shares, but only in *best effort*

Pricing the IPO

- ✓ Usually, IPOs are “underpriced”
 - prices **rise** significantly right after the stocks start trading in the secondary market
 - by about 17% on average in the US and UK
 - there are exceptions, like Facebook
 - Look (optional): “On May 18, 2012 Facebook opened for trading at 11:30 a.m. slightly above the offer price of \$38 per share. During the first sixty seconds of trading, over 158 million shares changed hands and the price ranged from a low of \$42 to a high of \$45. The high price for the day of \$45 happened within the first minute of trading.”
- ✓ This means that the issuer potentially leaves significant amounts of money “on the table”
- ✓ Markets (and issuing companies) have come to like underpriced IPOs as a *signal of “quality”* of IPO

Exercise: Pricing the IPO

Twitter's 70 M shares were initially sold to the public for \$26 per share (**offer price**), with underwriters retaining \$0.845 per share as fees and doing firm commitment underwriting.

Q1: What were the total proceeds from this IPO?

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Q2: How much did Twitter and its underwriters receive?

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Q2: How much did Twitter and its underwriters receive?

A2: Twitter received $70\text{M} \times (\$26 - \$0.845) =$
 $= 70\text{M} \times \$25.155 = \$1,760.85\text{M}$

The underwriter IBs got

- $70\text{M} \times \$0.845 = \59.15M
- $\$1.82\text{bn} - \$1,760.85\text{M} = \$59.15\text{M}$

Exercise: Pricing the IPO

By the end of the first trading day, Twitter's price rose to \$44.90 per share.

Q3: Was this IPO underpriced? If so, by how much?

Q4: What was Twitter's market capitalization at the end of the first day of trading, if the total number of Twitter's shares is 544,696,816?

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Q4: What was Twitter's market capitalization at the end of the first day of trading, if the total number of Twitter's shares is 544,696,816?

A4: The total market cap. of Twitter at the end of the first day is $544,696,816 \times \$44.90 = \$24,456,887,038$ (say \$ 24.46 bn)